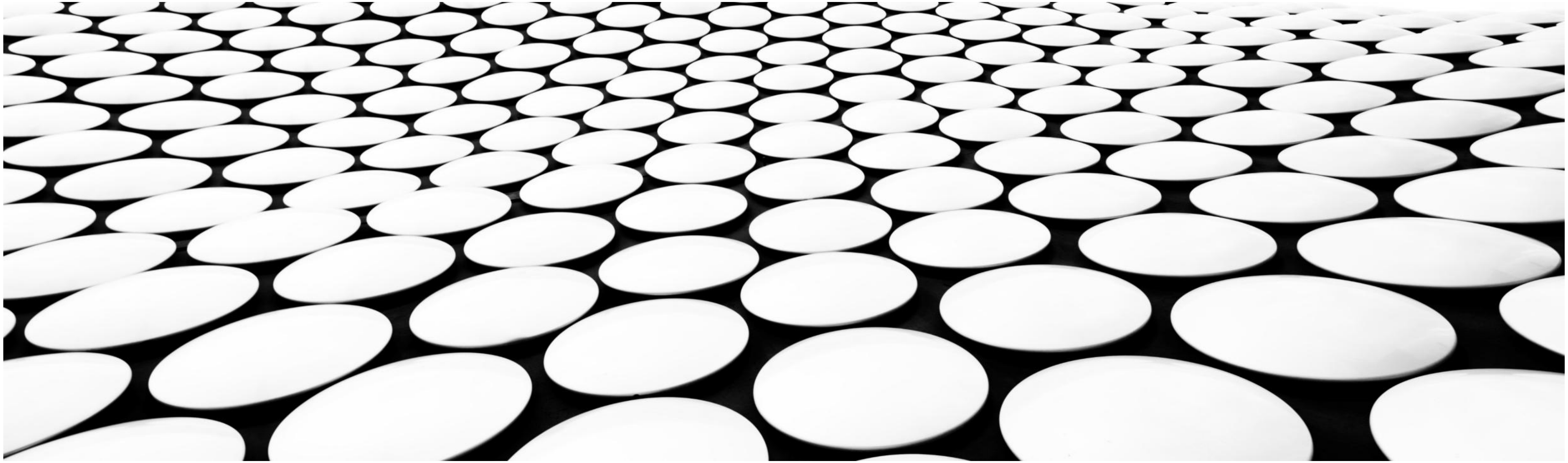
MISSING DATA

From identification to imputation



MISSING DATA

- Identification of missing data
- Types of missing data
- Removal approaches
- Imputation approaches
- MICE
- Evaluation

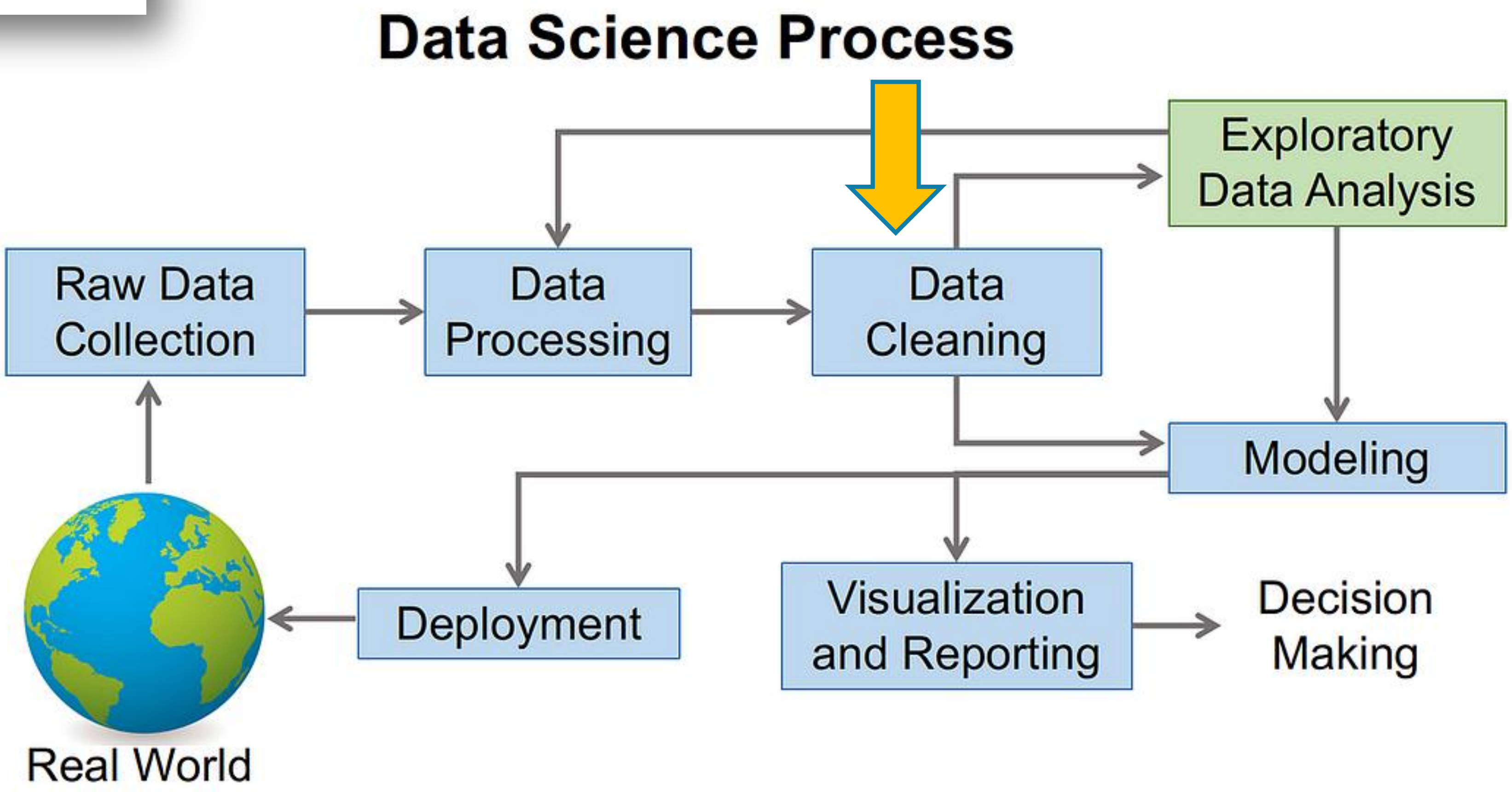
Today

Friday + lecture on Biases

Data Science Process: A Comprehensive Guide

Abhijit · Follow
6 min read · Jan 15, 2024

Source



DATA CLEANING – MISSING DATA

Detection → Understand → Fix

Could be expressed differently
Could be the result of previous validation steps

Forms it can take:
- MCAR missing completely at random
- MAR missing at random
- MNAR missing not at random

Removal
Imputation

Missing data

How to find it

Detection

Top Techniques to Handle Missing Values Every Data Scientist Should Know

Source

Functions	Descriptions
.isnull()	This function returns a pandas dataframe, where each value is a boolean value True if the value is missing, False otherwise.
.notnull()	Similarly to the previous function, the values for this one are False if either NaN or None value is detected.
.info()	This function generates three main columns, including the "Non-Null Count" which shows the number of non-missing values for each column.
.isna()	This one is similar to isnull and notnull. However it shows True only when the missing value is NaN type.

Read the dataset documentation as to the meaning of each…

Python functions

Detection

Titanic Data

```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
```

Census Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32561 entries, 0 to 32560
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             32561 non-null  int64
 1   work-class      30725 non-null  object
 2   fnlwgt          32561 non-null  int64
 3   education       32561 non-null  object
 4   education-num   32561 non-null  int64
 5   marital-status  32561 non-null  object
 6   occupation      30718 non-null  object
 7   relationship    32561 non-null  object
 8   race            32561 non-null  object
 9   sex             32561 non-null  object
 10  capital-gain    32561 non-null  int64
 11  capital-loss    32561 non-null  int64
 12  hours-per-week  32561 non-null  int64
 13  native-country  31978 non-null  object
 14  income          32561 non-null  object
```

Missing values are written differently in the data (need to find it)

Detection

ID	Name	Age	Height (cm)	Weight (kg)
1	Alice	25	165	55
2	Mohamed	null	170	-1
3	Mei	30	-1	62
4	Javier	22	180	75
5	Amina	28	160	58
6	Carlos	null	175	70
7	Rajesh	3[illegible]	-1	68
8	Yuki	40	162	65
9	Nia	27	172	-1
1[illegible]	[illegible]	[illegible]	178	80

Might come from outlier detection and replacement

No « range » check performed before, and take -1 as missing

Missing data

Understanding

Understand

Data Imputation: A Comprehensive Guide to Handling Missing Values

Ajay Verma · Follow
3 min read · Jul 14, 2024

Source

Types of Missing Values

1. MCAR (Missing Completely at Random)
2. MAR (Missing at Random)
3. MNAR (Missing Not at Random)

Let's start with an example

Some patients forgot to record their weight during a checkup, and their missing weight data is not related to other factors like age or health conditions.

MCAR

MAR

Patients who are older are less likely to report their smoking habits. The missingness of smoking data can be explained by the age of the patient, which is available.

Patients with severe health conditions may avoid answering questions about their disease because they feel uncomfortable or stigmatized. The missingness is related to the severity of the condition.

MNAR

MCAR (Missing Completely at Random):

MCAR occurs when the missing values are randomly distributed across the dataset, and there's no underlying pattern or correlation with other variables.

Example:
A survey respondent randomly skips a question.

Detection:
Compare the descriptive statistics (average, mean, std) of various features using subsets of data with and without the missing values. If similar then probably MCAR.

MCAR (Missing Completely at Random):

Patient Age	Exercise (hrs/week)	BMI	Blood Pressure
25	4	22	118
31	2	27	NaN
45	1	29	135
52	0	30	140
29	3	24	NaN
60	1	28	138
38	2	26	130

MAR (Missing at Random):

MAR occurs when the missing values are related to other observed variables, but not to the missing value itself.

Example:
A respondent’s not answering about whether they smoke or not based on their age.

Detection:
Try to count missing values based on other categories (split in categories, and count)

MAR (Missing at Random):

Age	Income (K$)	Education	Loan Amount (K$)
24	45	College	100
28	52	Bachelor	120
33	60	Bachelor	140
40	72	Master	180
52	NaN	Master	220
58	NaN	Bachelor	240
65	NaN	PhD	260

MNAR (Missing Not at Random):

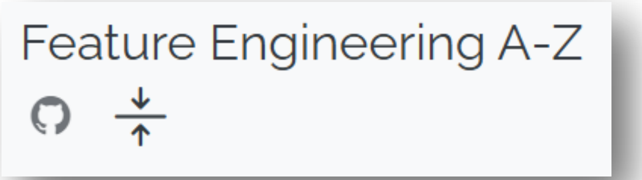
MNAR occurs when the missing values are related to the missing value itself, and not just to other observed variables.

Example:
A respondent’s monthly spending is missing because it’s too high or too low, and they didn’t want to disclose it.

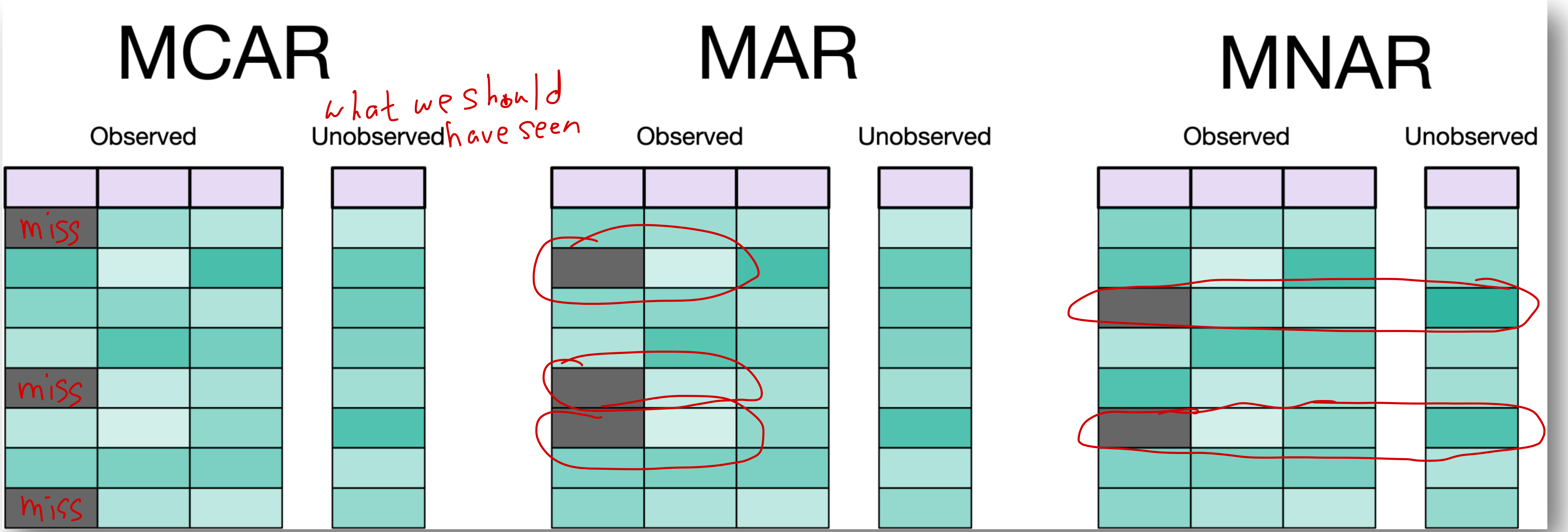
Detection:
Difficult to detect… not dependent on other categories. Perhaps use domain knowledge (expectations) or finding skewed distribution.

MNAR (Missing Not at Random):

Customer Age	Annual Spending ($)	Membership Type	Satisfaction Score
22	350	Basic	NaN
35	900	Gold	9
41	750	Silver	8
28	500	Basic	NaN
50	1200	Gold	10
31	650	Silver	8
45	980	Gold	9


Feature Engineering A-Z

Source


MCAR
Observed
Unobserved
what we should have seen
miss
miss
miss
MAR
Observed
Unobserved
MNAR
Observed
Unobserved

Missing data

Removal

Fix

Missing data → Removal, Imputation

Removal:

1) Listwise removal
2) Pairwise removal
3) Attribute removal

Listwise removal

```python
# Example in Python: Dropping rows with missing values
df.dropna(inplace=True)
```

In **listwise deletion**, any row (or observation) with **any missing value** in any of the variables you're analyzing is completely removed from the analysis.

Also called *observation removal* or *row removal*

Valid for MCAR, MAR and MNAR ??

MCAR

ID	Age	Weight	Height	Income
1	25	70	175	50000
2	30	80	180	60000
3	28	null	170	55000
4	null	65	160	null
5	22	60	165	45000

Pros and cons?

Resulting Data:

ID	Age	Weight	Height	Income
1	25	70	175	50000
2	30	80	180	60000
5	22	60	165	45000

Pairwise deletion

Leads to different subsets of the data usable for different analysis

In **pairwise deletion**, rows are removed **only for the variables** that are missing in a particular analysis. This means that a row may be included in one analysis (for some variables) but excluded in another (if those variables have missing data).

Student	Study Hours	Sleep Hours	Exam Score
A	5	7	80
B	6	NaN	85
C	NaN	6	78
D	8	5	92
E	7	8	NaN

Pros and cons?

Analysis: Correlation between Study Hours and Exam Score

Student	Study Hours	Exam Score
A	5	80
B	6	85
D	8	92

Student	Sleep Hours	Exam Score
A	7	80
C	6	78
D	5	92

Analysis: Correlation between Sleep Hours and Exam Score

Variable removal

If an attribute contains multiple missing values, we might wonder if that attribute would be usable at all for any predictive model, and remove it.

Also called *attribute removal* or *column removal*

Patient ID	Age	Exercise (hrs/week)	BMI	Blood Pressure
1	25	4	22	118
2	32	NaN	24	122
3	40	NaN	26	128
4	48	1	28	135
5	58	NaN	29	138
6	65	NaN	30	142
7	72	NaN	31	

Patient ID	Age	BMI	Blood Pressure
1	25	22	118
2	32	24	122
3	40	26	128
4	48	28	135
5	58	29	138
6	65	30	142
7	72	31	145

Pros and cons?

Missing data

Univariate imputation

Fix

Missing data → Removal / Imputation

1) Arbitrary value imputation
2) Default value imputation
3) Random Sample imputation
4) Time-based imputation

univariate

5) Conditional imputation
6) Correlational imputation

bivariate

7) Regression imputation
8) Predictive imputation
9) Similarity-based imputation

multivariate

requires predictive modeling

Arbitrary Value Imputation

Replace missing values with an arbitrary value.

Example:
Replace missing values in a categorical feature with a new category “Unknown”.

Of interest if the « missingness » carries meaning.

Customer	Age	Preferred Contact Method	Subscription Plan
A	30	Email	Premium
B	45	?	Standard
C	28	SMS	Premium
D	50	?	Basic

Pros and cons?

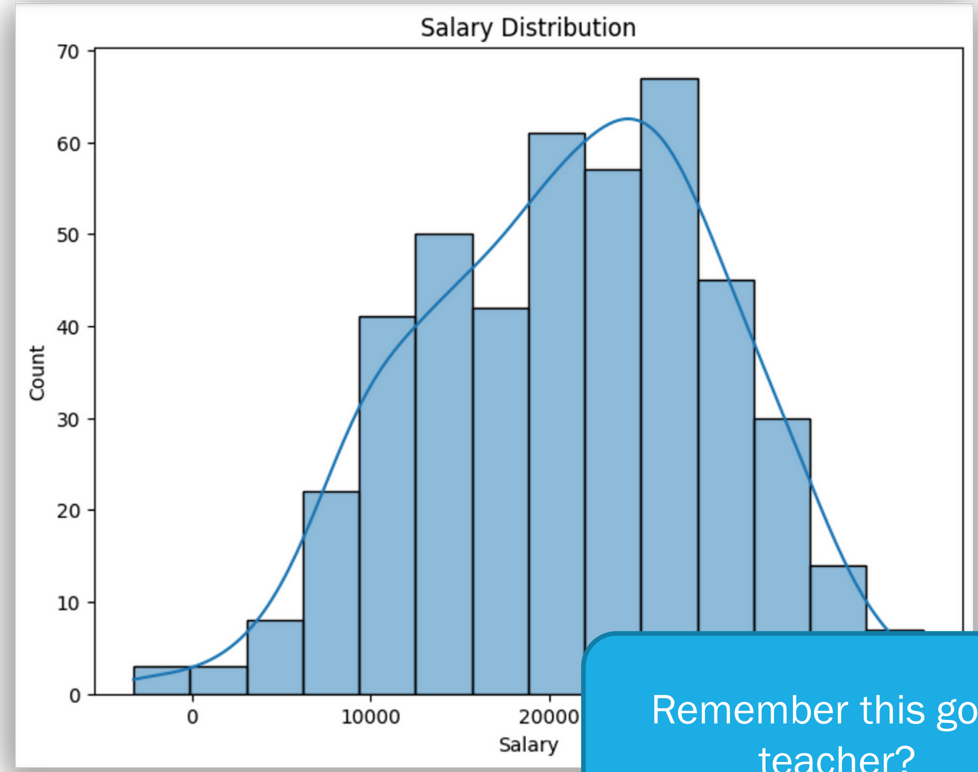
Customer	Age	Preferred Contact Method	Subscription Plan
A	30	Email	Premium
B	45	**Unknown**	Standard
C	28	SMS	Premium
D	50	**Unknown**	Basic

Default Value Imputation

Mean/Median/Mode Imputation: Replace missing values with the mean, median, or mode of the respective feature.

Example:
Replace missing values for the age feature with the average age in the dataset.

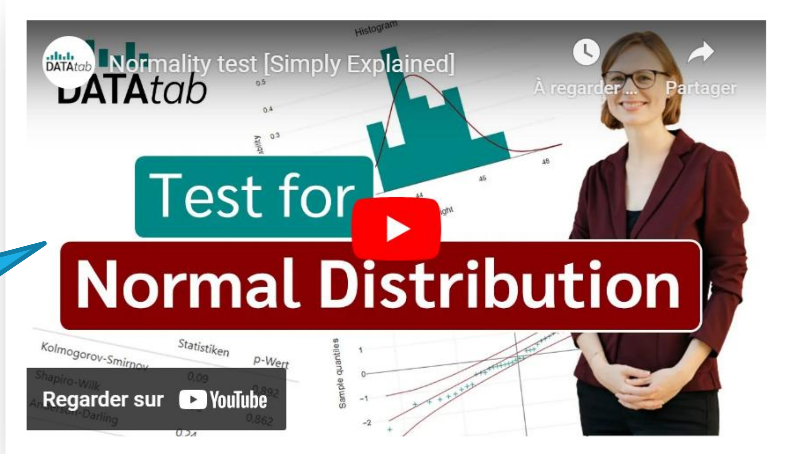
Before using such strategy, we should verify the distribution. Either visually… (less precise) or with some normality tests.



Remember this good teacher?

Shapiro-Wilk test

Shapiro-Wilk test is a hypothesis test that is applied to a sample with a null hypothesis that the sample has been generated from a normal distribution.



Source

Example of median imputation (numerical data)

Student ID	Age	Height
1	18	165
2	22	null
3	20	170
4	null	160
5	19	168

Student ID	Age	Height
1	18	165
2	22	166.5
3	20	170
4	19.5	160
5	19	168

Pros and cons?

Example of mode imputation (categorical data)

Customer	Age	Preferred Payment Method
A	30	Credit Card
B	45	PayPal
C	28	?
D	50	Credit Card
E	35	Credit Card
F	40	?

Customer	Age	Preferred Payment Method
A	30	Credit Card
B	45	PayPal
C	28	**Credit Card** (imputed)
D	50	Credit Card
E	35	Credit Card
F	40	**Credit Card** (imputed)

Pros and cons?

Random Sample Imputation

<u>Algorithm:</u>

- Create two subsets from the original data (D1, D2)
  - D1 contains all the observations without missing data
  - D2 contains those with missing data.
- Randomly select from each subset a random observation (R1, R2)
- R2's missing value is replaced by R1's value
- Repeat the process until there is no more missing information.

Student ID	Age	Height
1	18	165
2	22	null
3	20	170
4	null	160
5	19	168

Pros and cons?

Student ID	Age	Height
1	18	165
2	22	170
3	20	170
4	20	160
5	19	168

doesn't introduce new value

Time-based Imputation

The time series methods of imputation assume the adjacent observations will be like the missing data.

Attention to quickly changing data

Two approaches:

- LOCF – Last Observation Carried Forward
- NOCB – Next Observation Carried Backward

Used a lot in medical and financial domain

Example: Tracking blood pressure over days

Day	Blood Pressure (mmHg)
1	120/80
2	**Missing**
3	**Missing**
4	125/85
5	**Missing**
6	130/90

LOCF

Day	Blood Pressure (mmHg)
1	120/80
2	120/80
3	120/80
4	125/85
5	125/85
6	130/90

NOCB

Day	Blood Pressure (mmHg)
1	120/80
2	125/85
3	125/85
4	125/85
5	130/90
6	130/90

Missing data

Bivariate/Multivariate imputation

Fix

Missing data

Removal

Imputation

1) Arbitrary value imputation
2) Default value imputation
3) Random Sample imputation
4) Time-based imputation

univariate

5) Conditional imputation
6) Correlational imputation

bivariate

7) Regression imputation
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multivariate

Conditional Imputation

This method lets you fill missing values by referencing other variables, creating <u>specific rules.</u>

Example
Calculate the mean salary for those over 50 and use this to fill in missing values for others in the same age bracket.

Where do these « rules » come from?

Product ID	Price	Category	Brand
1	100	Electronics	Brand A
2	null	Apparel	Brand B
3	150	Electronics	Brand C
4	null	Accessories	Brand D
5	120	Electronics	Brand E
6	75	Apparel	Brand F

Pros and cons?

Product ID	Price	Category	Brand
1	100	Electronics	Brand A
2	75	Apparel	Brand B
3	150	Electronics	Brand C
4	75	Accessories	Brand D
5	120	Electronics	Brand E
6	75	Apparel	Brand F

Look at similar category

« Rule-based » use domain knowledge

Correlational Imputation

Use the correlation between two numerical features to impute missing values.

Example:
If a feature *DistanceFromWork* is correlated with a feature T*ransportTimeToWork*,
use their correlation to estimate missing values of T*ransportTimeToWork.*

- A common approach is using **linear correlation formulas:**

$$Y_{\text{missing}} = \rho \times \cancel{\frac{\sigma_Y}{\sigma_X}} \times (X - \mu_X) + \mu_Y$$

where:

$\rho(X - \mu) + \mu$

- ρ is the Pearson correlation coefficient. $= \rho X - \rho\mu + \rho\mu$
- μ and σ are the mean and standard deviation of X and Y.

$= \rho X$

If a strong correlation was found between math score and reading score

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

One possible correlation-based linear prediction

$$\text{Math Score (predicted)} = \mu_{\text{Math}} + r \times \frac{\sigma_{\text{Math}}}{\sigma_{\text{Reading}}} \times (\text{Reading Score} - \mu_{\text{Reading}})$$

Regression imputation

Use regression models to predict the missing values based on other features.

Example:
Build a regression model for house price prediction based on different attributes.

Use other variables to build a regression model

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

$$\text{Math Score} = \beta_0 + \beta_1 \times \text{Reading Score} + \beta_2 \times \text{Writing Score} + \beta_3 \times \text{Absences}$$

- $\beta_0 = 50$ (intercept)
- $\beta_1 = 0.2$ (coefficient for **Reading Score**)
- $\beta_2 = 0.3$ (coefficient for **Writing Score**)
- $\beta_3 = -1.5$ (coefficient for **Absences**)

These parameters are learned for « best fit ».

$$\text{Math Score} = \beta_0 + \beta_1 \times \text{Reading Score} + \beta_2 \times \text{Writing Score} + \beta_3 \times \text{Absences}$$

Size (sqft)	Location	Age (years)	Price ($)
1200	Urban	5	350K
1800	Suburban	10	NaN
1500	Urban	3	400K
2200	Rural	15	NaN
1400	Suburban	8	320K
1600	Rural	12	310K

What about categorical data in the model?

Size (sqft)	Age (years)	Price ($)	Location_Urban	Location_Suburban	Location_Rural
1200	5	350K	1	0	0
1800	10	NaN	0	1	0
1500	3	400K	1	0	0
2200	15	NaN	0	0	1
1400	8	320K	0	1	0
1600	12	310K	0	0	1

$$\text{Price} = \beta_0 + \beta_1 \cdot \text{Size} + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Location\_Urban} + \beta_4 \cdot \text{Location\_Suburban} + \beta_5 \cdot \text{Location\_Rural}$$

Predictive imputation

Train a predictive model (e.g. linear regression, logistic regression, decision trees, multilayer perceptron, etc) to predict missing values based on other features.

An expansion on regression imputation (more generic)

Instead of regression, could be any model for prediction. For example decision trees (we will discuss next week)

Similarity-based imputation

K-Nearest Neighbors (KNN) Imputation:

Find the k most similar rows to the one with missing values and impute the missing value based on the values of these neighbors.

To keep everything in feature space, we can do a one-hot encoding for categorical values

How KNN Imputation works

1. Set K
   - Decide on how many neighbors will be considered

2. **Finding Nearest Neighbors:**
   - Find the k closest rows (datapoint) based on a specific distance
   - Similarity is often determined using **Euclidean distance** in feature space.

3. **Averaging the Neighbors' Values:**
   - The missing values are replaced with the mean of the nearest neighbors.

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

The Euclidean distance between two students i and j is given by:

$$\text{distance}(i, j) = \sqrt{(R_i - R_j)^2 + (W_i - W_j)^2 + (A_i - A_j)^2}$$

Where:

- R is the **Reading Score**
- W is the **Writing Score**
- A is the **Absences**

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

Assume k=3

- **Student B** (distance = 9.49)
- **Student A** (distance = 9.64)
- **Student D** (distance = 15.81)

$$\text{Math Score (imputed)} = \frac{85 + 72 + 90}{3} = \frac{247}{3} \approx 82.33$$

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	82.33	80	85	7
D	90	92	94	2
E	65	70	68	4

Fix

Missing data

Removal

Imputation

1) Arbitrary value imputation
2) Default value imputation
3) Random Sample imputation
4) Time-based imputation

univariate

5) Conditional imputation
6) Correlational imputation

bivariate

7) Regression imputation
8) Predictive imputation
9) Similarity-based imputation

multivariate

Missing data

Multiple Imputation Chained Equations

Multiple Imputation by Chained Equations

Multiple Imputation by Chained Equations (MICE for short) is one of the most popular imputation algorithm in multivariate imputation.

MICE is an iterative algorithm which repeatedly performs imputation on each variable, one at a time, using the other variables to perform the prediction.

It can use any imputation approach.

How MICE works

1. **Initialize**: Use Default Value Imputation for initialization.

2. **Select a variable**: Choose a variable V to impute.

3. **Build a model**: Use the other variables in the dataset to build a regression (or any predictive) model that predicts V.

4. **Replace missing values**: Use the predictive model to predict and fill in the missing values for V.

5. **Repeat**: Repeat steps 2–4 for each variable with missing values.

Use the imputed values to repeat the whole process

Repeat until?

Patient	Age	Exercise (hrs/week)	BMI
1	25	4	22
2	32	NaN	24
3	40	2	NaN
4	48	1	28
5	58	NaN	29

Initialization of iterative procedure with average placeholders

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.33	24
3	40	2	25.75
4	48	1	28
5	58	2.33	29

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.33	24
3	40	2	25.75
4	48	1	28
5	58	2.33	29

Build a predictive model of Exercise (e.g. linear regression) using Age and BMI

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.8	24
3	40	2	25.75
4	48	1	28
5	58	1.5	29

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.8	24
3	40	2	25.75
4	48	1	28
5	58	1.5	29

Build a predictive model of BMI (e.g. linear regression) using Age and Exercise

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.8	24
3	40	2	26.2
4	48	1	28
5	58	1.5	29

Patient	Age	Exercise	BMI
1	25	4	22
2	32	2.8	24
3	40	2	26.2
4	48	1	28
5	58	1.5	29

Continue to alternate:

- Predict Exercise (model from Age, BMI)
- Predict BMI (model from Age, Exercise)

Ending criteria

- Fixed number of iterations
- Stability

Missing data

Evaluation

DATA CLEANING – MISSING DATA

Detection → Understand → Fix

Detection

Could be expressed differently
Could be the result of previous validation steps

Understand

Forms it can take:

- MCAR
- MAR
- MNAR

Fix

Removal
Imputation

As we take action, we wish to evaluate the impact

Correlation imputation
76.7

KNN imputation
82.33

Regression imputation
81

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

Student C was actually **80**. The formula for MSE is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_{\text{true},i} - y_{\text{pred},i})^2$$

Where:

- y_{true} is the actual (true) value.
- y_{pred} is the predicted (imputed) value.
- n is the number of predictions (in our case, only one student with a missing value).

Correlation imputation
76.7

KNN imputation
82.33

Regression imputation
81

Student	Math Score	Reading Score	Writing Score	Absences
A	85	88	90	5
B	72	75	78	3
C	?	80	85	7
D	90	92	94	2
E	65	70	68	4

$$MSE_{\text{correlation}} = (80 - 76.7)^2 = 3.3^2 = 10.89$$

$$MSE_{\text{KNN}} = (80 - 82.33)^2 = (-2.33)^2 = 5.43$$

$$MSE_{\text{regression}} = (80 - 81)^2 = (-1)^2 = 1$$

Evaluation Methods for imputation approaches

- **Cross-validation with simulation:** Evaluate imputation accuracy by simulating missing data and comparing imputed values to true values.

Simulated data

- Intentionaly remove data from the dataset (simulate either MCAR, MAR, MNAR)
- Run an imputation method
- Evaluate using metrics

Numerical Data

Larger errors are more penalized in MSE than MAE

Mean Square Error

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Categorical Data

Accuracy

$$Accuracy = \frac{\text{Number of correct imputations}}{\text{Total number of missing values}}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

Evaluation Methods for imputation approaches

- **Cross-validation with simulation:** Evaluate imputation accuracy by simulating missing data and comparing imputed values to true values.

- **Comparison to a baseline method:** Compare performance against simple methods like mean or median imputation.

Using same metrics, continue developing imputation approaches and validate that they get better

Evaluation Methods for imputation approaches

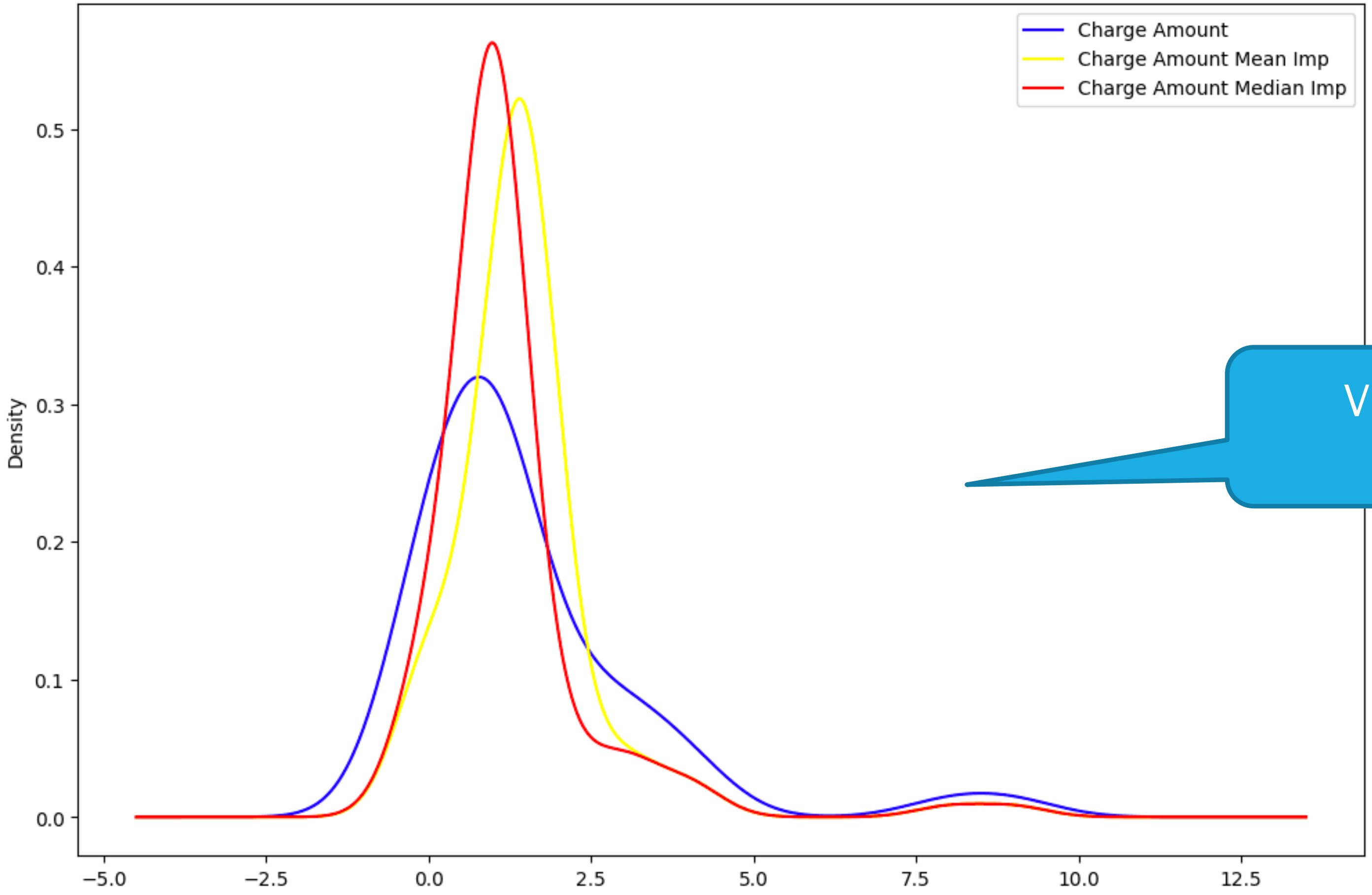
- **Cross-validation with simulation:** Evaluate imputation accuracy by simulating missing data and comparing imputed values to true values.

- **Comparison to a baseline method:** Compare performance against simple methods like mean or median imputation.

- **Visual inspection:** Use plots and graphs to visually check the plausibility of imputed data.

- **Distribution comparison:** Compare the distribution of imputed values to the distribution of non-missing data.

Distribution comparison



Visually done, or with statistical tests

Evaluation Methods for imputation approaches

- **Cross-validation with simulation:** Evaluate imputation accuracy by simulating missing data and comparing imputed values to true values.

- **Comparison to a baseline method:** Compare performance against simple methods like mean or median imputation.

- **Visual inspection:** Use plots and graphs to visually check the plausibility of imputed data.

- **Distribution comparison:** Compare the distribution of imputed values to the distribution of non-missing data.

- **Consistency with domain knowledge:** Ensure that imputed values are reasonable and make sense contextually.

Validate with domain expert

Evaluation Methods for imputation approaches

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- **Impact on model performance:** Evaluate how imputation affects the performance of downstream models.

Construct predictive model and validate results with various imputation methods – very costly

MISSING DATA

- Identification of missing data
- Types of missing data
- Removal approaches
- Imputation approaches
- MICE
- Evaluation

Today

Friday + lecture on Biases